Data Science and AI for Finance
DA6701

Session 1 – Course Overview

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Course Overview

- Course Objective and references
- Course Outline/ walkthrough
- Grading and TAs

Course objective and references

- Objective: Finance fundamentals → Basic mathematical and probabilistic constructs → Computational Intelligence
- This course provides an overview of financial basics, which is followed by a comprehensive framework that bridges the financial math with Machine Learning approaches.
- Style: Finance 101 heavy at the start, micro-case centric (who is suited; poll)
- Pre-reqs: PRML/FML, MFDS, Data Analytics Lab (no finance course)
- References:

•**Bodie, Z., Kane, A., & Marcus, A. J. (2014).** *Investments* (10th ed.). New York: McGraw-Hill Education [1:7, 1:11].
•**Jansen, S. (2020).** *Machine Learning for Algorithmic Trading: Predictive models to extract signals from market and alternative data for systematic trading strategies with Python* (2nd ed.). Birmingham, UK: Packt Publishing [4:3882, 4:3884].
•**Hull, J. C. (2022).** *Options, Futures, and Other Derivatives* (11th Global ed.). Harlow, England: Pearson Education Limited [3:2326, 3:2327].
•**Capiński, M., & Zastawniak, T. (2003).** *Mathematics for Finance: An Introduction to Financial Engineering*. London: Springer-Verlag [2:1842, 2:1843].

Course outline as modules

1. Finance fundamentals: concepts, environments, products, pricing
2. Investment and Portfolio Management
3. Risk Management and Compliance
4. Trading and Market Analysis
5. Credit and Consumer Insights
6. Contemporary topics

Course walkthrough

1. Finance Fundamentals: concepts, environments, products and pricing
   a. Concepts: Basics of compounding (log returns and stock price modelling); EMH theory, no arbitrage principal and one-step binomial tree; risk and expected return
   b. The financial environment: Overview of markets and Financial instruments
   c. Basics of Asset valuation and pricing. Covers stocks, futures, Options, ETFs, bonds, credit/loans, and insurance
   d. Basics of data types and feature engineering.
   e. Cases: Stock valuation and Insurance modeling.
2. Investment and Portfolio Management
   a. Modeling risky asset dynamics
   b. Portfolio optimization
   c. Quantitative Asset Pricing
   d. Personal finance
   e. Cases: Optimal and replicating portfolio, Personal portfolio management, annuities

Course Walkthrough

3. Risk and compliance
    a. Risk Management (VaR, etc).
    b. Stress testing and simulation
    c. Fraud detection
    d. Financial Operations and automation
    e. cases: Insider trading and front running case study
4. Trading and Market Analysis
    a. Overview of how a financial market works and sell-side operations
    b. Market mechanisms (stakeholders), order types and data.
    b. Arbitrage and trading strategies (pipeline and specific strategies)
    d. cases: more stock prediction, order flow to order book, order book prediction, regularization for hedging, volume/volatility prediction, TWAP/VWAP, RL for trading, sentiment analysis/social media.

Course Walkthrough

5. Credit and consumer insights

    a. Delinquency Modelling (Point of Entry vs Point in Time, class imbalance problems, individual vs cohort, induction vs transduction)

    b. Consumer choice models with multilabel classification

    c. customer segmentation with clustering algorithms

    d. Cases: Three cases on delinquency modelling, customer segmentation, and customer choice

6. Contemporary topics

    a. Ethics in finance

    b. Quantum finance

    c. Blockchains in finance

    d. Case showing impact of rule based decision engines and feedback loops.

Grading and TA

- Micro case assignments
  - Groups of 4-5 students
  - Each assignment will need you to work with data using Python to achieve a certain task (predict something, group customers, minimize risk, etc.)
  - Each assignment will have a 3 page submission
    - First page outlines the process you followed (how did you process the data? what is the flow chart you followed?)
    - Second page show cases results you achieved (not an output of code but in a readable report format)
    - Third page is like an appendix that shows snapshots of code and results directly.
  - Each assignment will require you to also submit the code which should execute to the results seen in the third page.
- Weekly class plan (no Tuesday theory)
- Grading
  - 70% in assignments
  - 30% End sem
- TAs